Defensive KYC: Presentation Attack Detection via Biologically-Anchored Telemetry and Spatio-Temporal Forensics

Manmohan Vishwakarma Ankur Raj
Department of Electronics & Communication Engineering
National Institute of Technology Patna, Bihar, India
April 2026

https://github.com/mv35011/AuthKYC--END-to-END-system-for-bank-kyc-against-AI-attacks

***Abstract*—Remote Know Your Customer (KYC) verification is the standard digital identity onboarding layer for banking and financial services. However, video-based KYC is fundamentally vulnerable to modern attack vectors: real-time deepfake pipelines (e.g., DeepFaceLab, FaceSwap), virtual-camera injection (OBS Studio, ManyCam), and screen-replay presentation attacks can impersonate a legitimate customer without triggering standard liveness checks.**

This paper presents AuthKYC, an end-to-end Presentation Attack Detection (PAD) system operating on live video streams. The core architectural insight is that synthetic generators optimise against *human visual perception*; they have no incentive to synthesise a coherent cardiovascular pulse, a consistent sensor-noise fingerprint, or plausible GAN-free frequency spectra. AuthKYC exploits these blind spots simultaneously through a modular 4-stage cascading security waterfall comprising hardware forensics, frequency analysis, biological liveness verification, and spatio-temporal deep learning.

On a mixed-domain evaluation dataset ($N = 21$), the system achieved 95.2% overall accuracy with 0% False Positive Rate (zero attacks bypassed the pipeline), outperforming a standard Xception baseline (46.92%) by a factor of $2\times$.

***Index Terms*—Presentation Attack Detection; PRNU Forensics; Moiré Pattern Analysis; Remote Photoplethysmography; 3D Convolutional Networks; Cross-Attention; Deepfake Detection; KYC Security**

I. Introduction

A. Problem Statement

Digital banking onboarding relies on video-based identity verification where a customer presents their face to a camera. This process is vulnerable to four distinct attack classes, summarised in Table I. No single detection technique addresses all four simultaneously: a replay detector cannot catch virtual camera injection; a liveness detector cannot catch a face-swap worn by a real human; and an AI deepfake detector trained on spatial features alone cannot distinguish a high-definition screen recording from a live camera feed.

B. Proposed Solution

AuthKYC introduces a **4-layer waterfall architecture** where each layer exploits a fundamentally different physical or biological constraint:

TABLE I: Attack Classes and Failure Modes of Single-Layer Detectors

Attack Class	Vector	Why Standard Detectors Fail
Virtual Camera Injection	OBS/ManyCam pipes video into OS driver	No screen artefacts; replay detectors see a “clean” feed
Screen Replay	Phone/tablet screen held to camera	Single-frame AI cannot distinguish screen pixels from sensor pixels
Live Face-Swap	Real human wearing real-time AR deepfake	Real pulse exists, bypassing basic liveness
Fully Synthetic Video	GAN/Diffusion talking head	Spatial detectors miss temporal inconsistencies

1) **S1 — PRNU Sensor Forensics:** Verifies that the video originates from a physical CMOS sensor via Photo-Response Non-Uniformity (PRNU) noise analysis.
2) **S2 — Moiré Frequency Analysis:** Detects screen-replay attacks by identifying periodic high-frequency Moiré interference in the 2D Fourier domain.
3) **S3 — rPPG Biological Liveness:** Extracts the sub-visible cardiovascular pulse from facial skin chrominance using the CHROM algorithm.
4) **S4 — FTCA Spatio-Temporal AI:** A novel architecture fusing a 3D-CNN with a differentiable frequency encoder through multi-head cross-attention.

The pipeline operates as a strict waterfall: failure at any stage triggers immediate session rejection, minimising unnecessary GPU computation.

II. System Architecture

A. Stage 1: PRNU Sensor Forensics

Physical Principle. Every CMOS image sensor contains manufacturing imperfections at the sub-pixel level — inhomogeneities in silicon doping and oxide thickness — that create a unique, persistent noise pattern called Photo-Response Non-Uniformity (PRNU). Virtual cameras (OBS, ManyCam) render frames entirely in software and carry *zero* physical sensor signature.

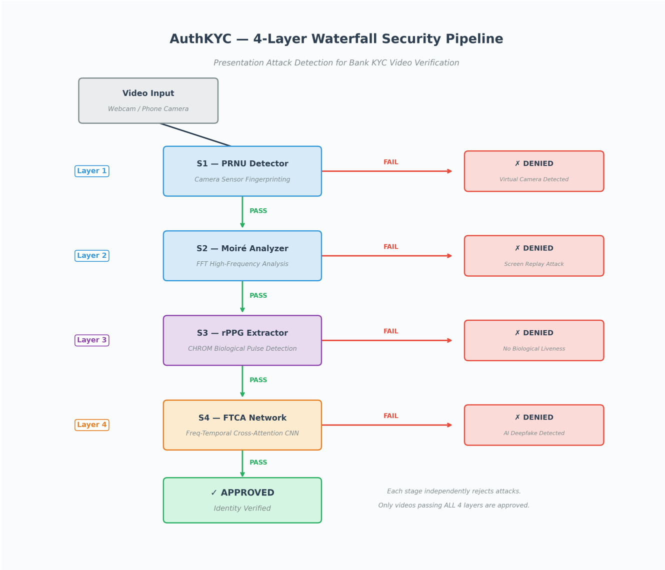

Fig. 1: AuthKYC 4-stage waterfall architecture. Each stage is designed to catch a distinct attack class; failure at any stage terminates evaluation.

Implementation. Each frame is converted to grayscale and downsampled to 480 px width. A median blur (kernel size 3; size 5 was empirically found to destroy real PRNU signal) produces the denoised estimate. The noise residual $r = \mathrm{int16}(\mathrm{orig}) - \mathrm{int16}(\mathrm{denoised})$ is accumulated over ≈ 180 frames; by the Law of Large Numbers, random shot noise cancels while the persistent fingerprint K survives. Two metrics are derived: *Noise Energy* (variance of K) and *Spectral Flatness*:

$$\mathrm{SF} = \frac{\exp\left(\frac{1}{N}\sum_i \ln|P_i|\right)}{\frac{1}{N}\sum_i |P_i|}$$

where P_i are 2D FFT power-spectrum values. Real PRNU has broadband noise (SF $\approx$ 0.4–0.8); compression block artefacts yield SF $\approx$ 0.1–0.3. Decision: $\mathrm{energy} > 0.5$ AND $\mathrm{SF} > 0.3 \Rightarrow$ physical camera confirmed.

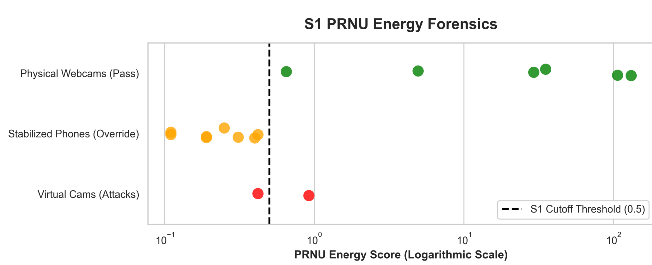

Fig. 2: PRNU energy distribution (log scale). Physical webcams (4.94–131.16) are clearly separated from virtual cameras and stabilised phones (0.11–0.92). The threshold at 0.5 cleanly divides both populations.

Physical webcams scored 4.94–131.16, while virtual cameras and screen replays scored 0.42–0.92. Stabilised phone cameras scored 0.11–0.42 due to computational photography pipelines destroying sensor noise, motivating the Dynamic Fallback (Section II-E).

B. Stage 2: Moiré Replay Detection

Physical Principle. When a video displayed on a screen is captured by a camera, the two regular grids create beat frequencies known as Moiré interference — periodic high-frequency spikes in the 2D Fourier magnitude spectrum absent in natural camera captures. Screen replays also *lose* high-frequency texture through double compression.

Implementation. After grayscale conversion and 640 px resizing, a 2D FFT is computed and zero-frequency shifted to centre. The central 15% of the spectrum is masked (to remove natural low-frequency content). High-frequency energy and a *peak ratio* $\mathrm{P}_{99}/\mathrm{median}$ are computed. Screen grids create sharp periodic peaks (ratio $\gg 5$), while natural scenes follow an approximate $1/f$ power-law decay (ratio $\approx$ 2–4). If peak ratio > 5, the anomaly score is boosted by $\mathrm{peak\_ratio}/3$. Score below 1500 $\Rightarrow$ classified as screen replay.

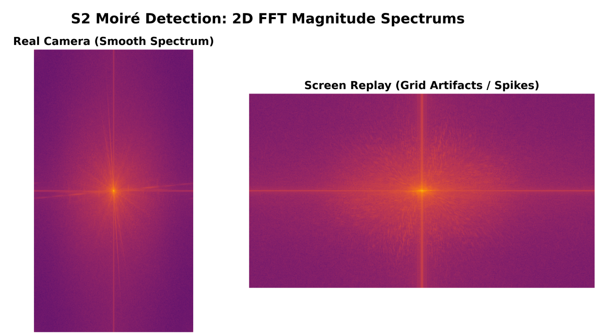

Fig. 3: 2D FFT magnitude spectra. *Left:* real camera capture shows smooth $1/f$ decay. *Right:* screen replay exhibits radial high-frequency spikes from Moiré interference between screen subpixels and camera sensor pixels.

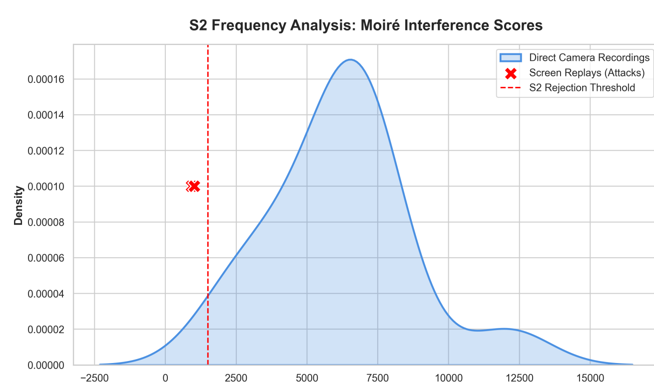

Fig. 4: Moiré score distribution. Screen replays cluster at 920–1025; all real camera recordings score above 2,041. The decision boundary at 1500 provides complete mathematical separation.

Screen replay videos scored 920–1025; real camera recordings scored 2,041–12,135. The threshold at 1500 provides **complete mathematical separation** with zero overlap.

C. Stage 3: rPPG Biological Liveness

Physical Principle. Living human skin exhibits sub-visible colour fluctuations driven by the cardiac cycle: haemoglobin absorption modulates reflected light intensity in the green channel ($\approx 540\,\mathrm{nm}$). Deepfake generators have no incentive to synthesise a coherent cardiovascular pulse waveform.

Implementation — CHROM Algorithm [4]. MediaPipe Face Mesh (468 landmarks) isolates three skin ROIs (forehead, right cheek, left cheek) via convex hull masks. Per-frame mean R, G, B is extracted; each channel is divided by its temporal mean (AC/DC separation) and linearly detrended to remove auto-exposure drift. The CHROM projection isolates the pulsatile blood volume component:

$$X = 3R_n - 2G_n - B_n \tag{1}$$

$$Y = 1.5R_n + G_n - 1.5B_n \tag{2}$$

$$\text{pulse} = X - \frac{\sigma_X}{\sigma_Y} \cdot Y \tag{3}$$

A 4th-order Butterworth IIR filter (0.7–4.0 Hz; 42–240 BPM) is applied via `filtfilt` for zero-phase distortion. Welch's PSD identifies the dominant frequency; SNR is computed as peak power divided by noise floor. Gate logic: $\mathrm{SNR} \geq 1.5\,\mathrm{dB}$ AND $45 \leq \mathrm{BPM} \leq 150 \Rightarrow$ biological liveness confirmed.

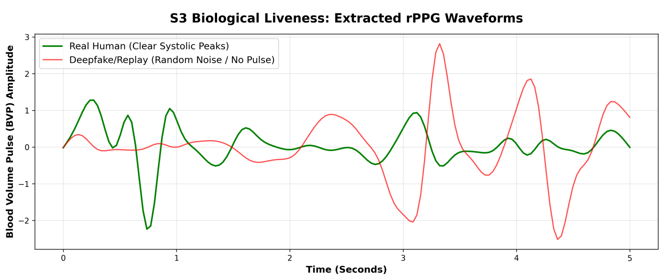

Fig. 5: Extracted Blood Volume Pulse (BVP) waveforms. *Top:* real human showing periodic systolic peaks at $\approx$75 BPM. *Bottom:* deepfake video yielding only chaotic noise with no discernible cardiac rhythm.

D. Stage 4: FTCA — Frequency-Temporal Cross-Attention

Motivation. Existing detectors operate either spatially (Xception, EfficientNet) or temporally (R3D, SlowFast), but not both. AuthKYC introduces a novel architecture that fuses RGB spatio-temporal features with frequency-domain features through cross-attention, compelling the model to learn *which* frequency artefacts correlate with *which* spatial-temporal movements.

RGB Branch. Input $[B, 3, 16, 224, 224]$ (16 contiguous MTCNN-cropped frames) is processed by a pretrained R3D-18 [5] through 4 residual layers of 3D convolutions, producing output $[B, 512, 2, 7, 7]$ flattened to a sequence $[B, 98, 512]$.

Frequency Branch. Per-frame grayscale conversion via luminance weights $(0.2989R + 0.5870G + 0.1140B)$ feeds a differentiable 2D DFT (`torch.fft.fft2`, orthonormal normalisation). The log-magnitude of the shifted spectrum $\log(|\mathrm{fftshift}(\mathcal{F})| + 10^{-8})$ is encoded by a 4-layer Conv3D encoder (1→64→128→256→512 channels, matching R3D-18's 8Œ temporal downsampling), producing $[B, 98, 512]$.

Cross-Attention Fusion.

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{64}}\right) V$$

where Q = RGB features, $K{=}V$ = frequency features; 8 attention heads, $d_{\mathrm{embed}} = 512$. A residual connection with LayerNorm follows: $\mathrm{out} = \mathrm{LN}(f_{\mathrm{RGB}} + \mathrm{attn\_out})$.

Classifier. AdaptiveAvgPool1d(1) → Linear(512→128) → ReLU → Dropout(0.5) → Linear(128→1). Sigmoid output $> 0.50 \Rightarrow$ deepfake.

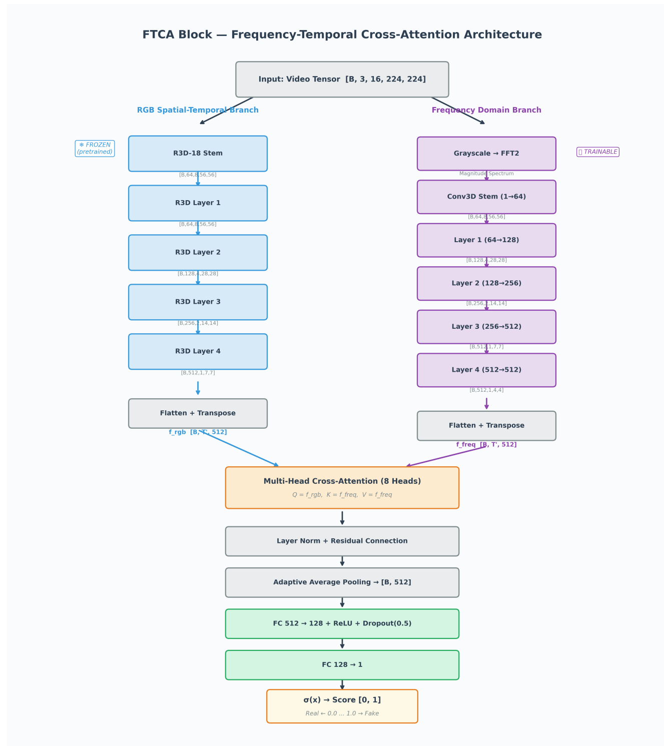

Fig. 6: FTCA block diagram. RGB and frequency branches process the same 16-frame clip in parallel; multi-head cross-attention fuses their representations before the binary classifier.

E. Dynamic Fallback Routing

Modern smartphones apply aggressive Electronic Image Stabilisation (EIS) and temporal denoising that destroy the S1 PRNU fingerprint. Without mitigation, every legitimate phone KYC session would be falsely rejected. The solution: if S1 fails but the system detects a verified biological heartbeat (S3 pass) AND zero AI manipulation (S4 pass), biological and AI evidence overrides the hardware failure:

```
if not s1_prnu_pass and s3_rppg_pass and
    s4_ftca_pass:
    s1_pass = True   # Dynamic Override
```

An attacker must simultaneously fake a biological pulse *and* pass the FTCA deepfake detector to exploit this override, ensuring security is not compromised.

III. Training Pipeline

A. Dataset and Extraction

Training data comprised FaceForensics++ C23 [1] (original + 5 manipulation methods: Deepfakes, Face2Face, FaceSwap, FaceShifter, DeepFakeDetection) and Celeb-DF v2 [2]. A balanced 1500 real + 1500 fake split (seed=42, 80/20 train/val) was used. MTCNN [7] detected and cropped faces to 224×224 (40 px margin); up to 8 contiguous 16-frame sequences were extracted per video.

B. Phase 1: Full FTCA Training

TABLE II: Phase 1 Training Hyperparameters (Full FTCA)

Parameter	Value
Architecture	FTCABlock (R3D-18 + FreqEncoder + CrossAttn)
Parameters	≈12.8M
Optimiser	AdamW (lr=10^{-4}, wd=10^{-3})
Scheduler	ReduceLROnPlateau (factor=0.5, patience=3)
Loss	BCEWithLogitsLoss
Precision	Mixed (AMP + GradScaler)
Batch Size	16
Epochs	30
Hardware	NVIDIA RTX A6000 (48 GB) via RunPod
Best Val Acc	**64.67%**
Best Val Loss	**0.6053**

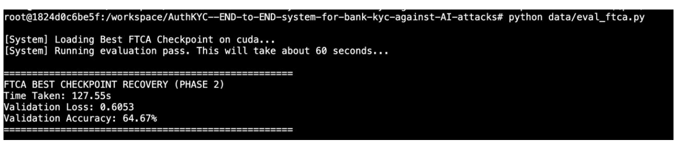

Fig. 7: Phase 1 training curves: validation accuracy and loss across 30 epochs.

C. Phase 2: Xception Baseline

To establish that temporal analysis is necessary, we trained an Xception [6] baseline using the `timm` library (≈22M parameters, pretrained ImageNet, Adam lr=10^{-4}, 20 epochs, best val accuracy: 46.92%). Xception's validation loss diverged from training loss after epoch 5 — classic overfitting to spatial compression artefacts rather than genuine deepfake features.

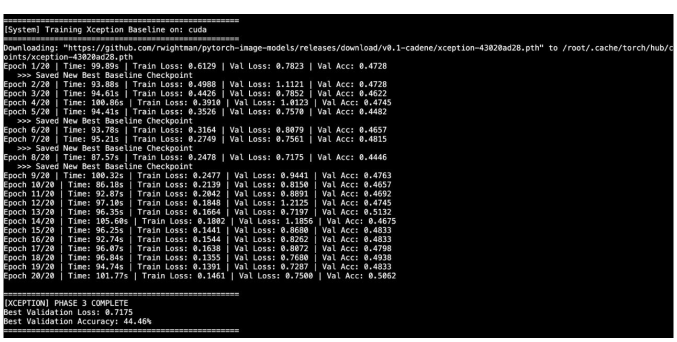

Fig. 8: Xception baseline training curves showing clear overfitting after epoch 5.

D. Phase 3: Domain Adaptation Fine-Tuning

To close the domain gap between FF++ training data and real-world webcam/phone footage, targeted fine-tuning was performed on a custom dataset: 250 webcam anchor videos (real recordings augmented via HFlip, ColorJitter, GaussianBlur), 250 FF++ original videos, and 500 FF++ fake videos. The entire R3D-18 backbone was frozen; only the FrequencyEncoder, CrossAttention, LayerNorm, and Classifier (≈5.2M parameters) were trained (AdamW lr=10^{-5}, 15 epochs).

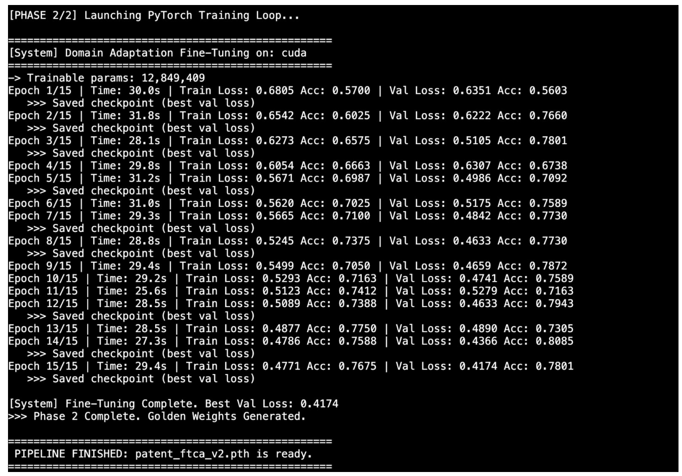

Fig. 9: Phase 3 fine-tuning curves. Domain adaptation improves validation accuracy from 64.67% to 80.85%.

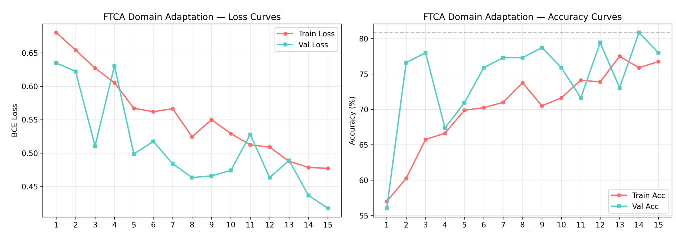

Fig. 10: Aggregated FTCA training and fine-tuning curves.

IV. Experimental Results

A. End-to-End Waterfall Evaluation (Experiment 6)

Dataset ($N = 21$): 12 real human videos (4 FF++ C23 originals + 8 uncompressed mobile recordings under varying lighting), 2 physical screen-replay attacks (phone screen held to MacBook webcam), and 7 FF++ C23 deepfakes (Deepfakes and Face2Face methods). Including compressed FF++ "Real" videos alongside uncompressed mobile recordings was a deliberate experimental control: if FTCA were learning the domain gap rather than genuine artefacts, these videos would have been misclassified. Their successful passage confirms the model evaluates spatio-temporal blending artefacts.

False Positive Rate: 0% — no attack of any type bypassed the waterfall.

Defence-in-Depth. Table V shows which stage intercepted each attack. A critical finding: screen replay attacks scored as

TABLE III: End-to-End Waterfall Results ($N = 21$)

Category	Total	Correct	Accuracy
Real Humans (Approval Rate)	12	11	91.7%
Screen Replays (Detection)	2	2	100.0%
Deepfakes FF++ (Detection)	7	7	100.0%
Overall	**21**	**20**	**95.2%**

TABLE IV: Confusion Matrix

	Pred. PASS	Pred. REJECT
Actual Real	TP = 11	FN = 1
Actual Attack	FP = 0	TN = 9

"real" by S4 (they contain genuine face textures), confirming that the physical S2 layer is indispensable.

TABLE V: Per-Stage Attack Attribution

Stage	Attacks Caught	Attack Types
S2 (Moiré)	3	2 replays + 1 compressed deepfake
S3 (rPPG)	1	Deepfake with no detectable pulse
S4 (FTCA)	5	Software deepfakes passing S2/S3
Escaped	0	—

B. Signal-Level Visualisations

Figure 13 shows FTCA score distributions: real videos score 0.017–0.294 (one outlier at 0.781); deepfakes score 0.614–0.800; screen replays score 0.024–0.040 (correctly classified by S2 *before* reaching S4).

C. Model Comparison

TABLE VI: Model Comparison

Model	Val Acc	Val Loss	Params	Temporal
FTCA (fine-tuned)	**80.85%**	**0.4174**	12.8M	Yes
FTCA (pretrained only)	64.67%	0.6053	12.8M	Yes
Xception Baseline	46.92%	0.6940	≈22M	No

D. Latency Analysis

End-to-end latency averaged 5.2 s for real webcam sessions, 14.1 s for phone inputs (higher resolution → more MTCNN processing), 8.5 s for replays, and 7.7 s for deepfakes. The primary bottleneck is the CPU-bound MTCNN face cropper and the mandatory 6-second rPPG biological buffer (≈180 frames at 30 fps). Planned optimisations — RetinaFace or YOLOv8-face and TensorRT INT8 quantisation — are projected to reduce latency below 3 s.

V. Dataset Composition and Domain Shift

C23 compression destroys the microscopic PRNU sensor fingerprint, so the 4 FF++ "Real" videos likely triggered the Dynamic Fallback at S1. However, C23 compression is light enough that the macro-pixel chrominance shifts of blood flow

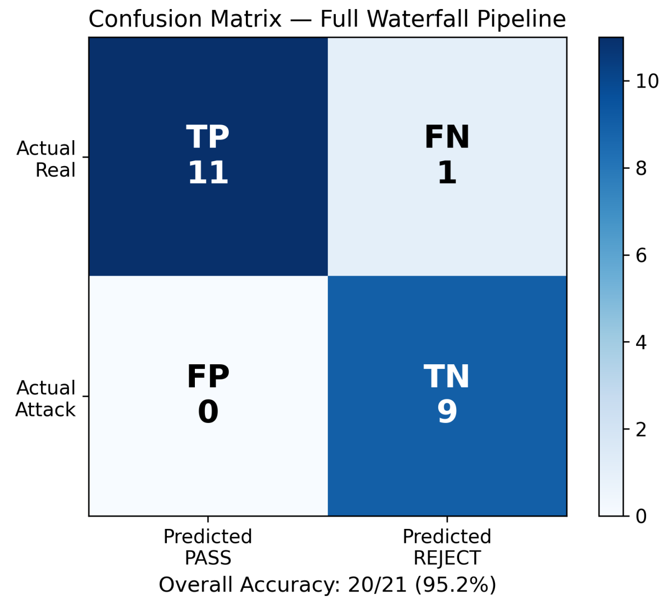

Fig. 11: Confusion matrix on the 21-video mixed-domain evaluation set.

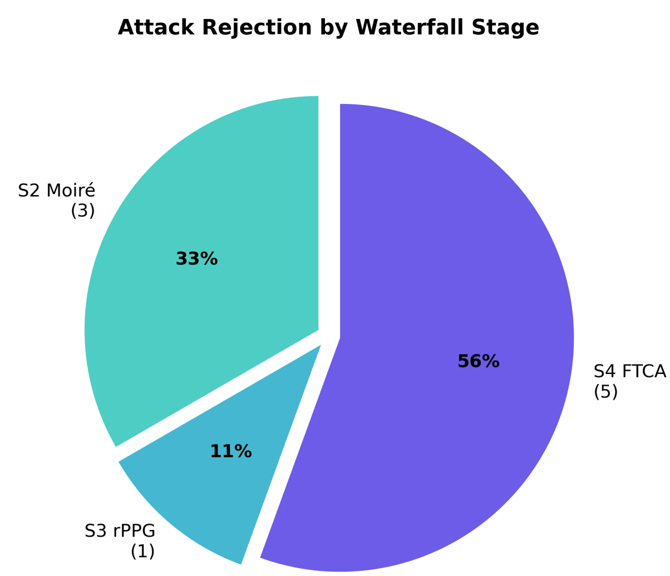

Fig. 12: Waterfall rejection diagram showing which stage intercepted each attack class.

survive S3 rPPG extraction. Their successful passage through S4 confirms the FTCA model evaluates genuine deepfake artefacts rather than compression-domain shortcuts — a cross-domain control validating that the system is not fitting the gap between compressed and uncompressed media.

VI. Limitations and Future Work

1) **Sample Size.** End-to-end validation was conducted on $N = 21$ videos, establishing proof-of-concept viability. Large-scale benchmarking requires institutional uncom-

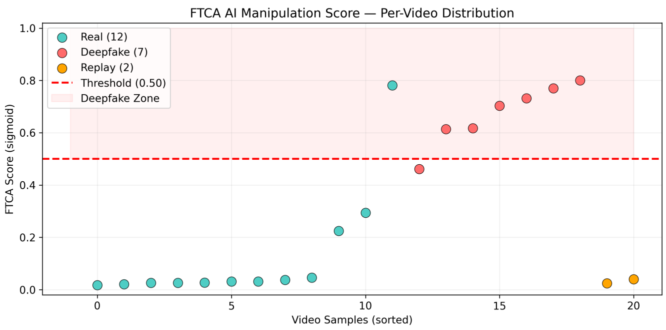

Fig. 13: FTCA sigmoid score distributions. Real and deepfake populations are well-separated around the 0.50 decision threshold. Screen replays score near zero (genuine textures), confirming S2 must precede S4.

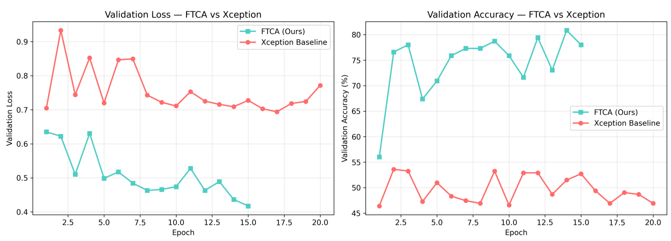

Fig. 14: FTCA vs Xception validation accuracy comparison across training epochs.

pressed biometric datasets (OULU-NPU, UBFC-rPPG, SiW).

2) **FTCA Accuracy.** 80.85% validation accuracy is strong for a minor project but below production thresholds; more training data and longer training schedules are required.
3) **Latency.** 5–15 s per video is acceptable for batch processing but too slow for real-time UX. TensorRT quantisation and faster face detectors are planned.
4) **PRNU on Phones.** Modern computational photography pipelines (HDR+, Night Sight, EIS) destroy PRNU signatures. The Dynamic Fallback partially addresses this, but a dedicated mobile PRNU model would be more robust.
5) **Adversarial Robustness.** The system has not been evaluated against perturbations specifically designed to fool individual pipeline stages.

VII. Conclusion

AuthKYC demonstrates that single-layer deepfake detection is fundamentally insufficient for securing banking KYC workflows. By fusing four orthogonal detection principles — hardware physics (PRNU), frequency-domain optics (Moiré), biological signals (rPPG), and learned spatio-temporal features (FTCA) — the system forces attackers to simultaneously satisfy physical, biological, and digital constraints. The 95.2% accuracy with zero false positives on a mixed-domain dataset, combined with a 2× improvement over the Xception spatial baseline, validates this multi-modal cascaded approach as a viable foundation for production PAD systems.

VIII. Code and Artifact Availability

The complete source code, training scripts, evaluation pipeline, and module implementations are publicly available:

- **Source Code (GitHub):** https://github.com/mv35011/AuthKYC--END-to-END-system-for-bank-kyc-against-AI-attacks
- **Trained Weights, Plots, and Artifacts (Google Drive):** https://drive.google.com/drive/folders/1ewJDwSbeyMI3JFJkYBhZD_NDlN92H-8w?usp=sharing

The repository contains: `modules/` (all 4 security stages), `core_engine.py` (orchestrator), `data/` (training scripts), `finetune/` (domain adaptation), and `run_experiment6.py` (ablation study). Pre-trained model weights (`patent_ftca_v2.pth`) and all report figures are available on the Drive link.

References

[1] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. NieSSner, "FaceForensics++: Learning to Detect Manipulated Facial Images," in *Proc. IEEE ICCV*, 2019, pp. 1–11.

[2] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A Large-scale Challenging Dataset for DeepFake Forensics," in *Proc. IEEE CVPR*, 2020, pp. 3207–3216.

[3] J. Lukáš, J. Fridrich, and M. Goljan, "Digital Camera Identification from Sensor Pattern Noise," *IEEE Trans. Inf. Forensics Security*, vol. 1, no. 2, pp. 205–214, 2006.

[4] G. de Haan and V. Jeanne, "Robust Pulse Rate From Chrominance-Based rPPG," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 10, pp. 2878–2886, 2013.

[5] D. Tran, H. Wang, L. Torresani, J. Ray, Y. LeCun, and M. Paluri, "A Closer Look at Spatiotemporal Convolutions for Action Recognition," in *Proc. IEEE CVPR*, 2018, pp. 6450–6459.

[6] F. Chollet, "Xception: Deep Learning with Depthwise Separable Convolutions," in *Proc. IEEE CVPR*, 2017, pp. 1251–1258.

[7] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks," *IEEE Signal Process. Lett.*, vol. 23, no. 10, pp. 1499–1503, 2016.